# EE5907 Pattern Recognition / EE5027 Statistical PR

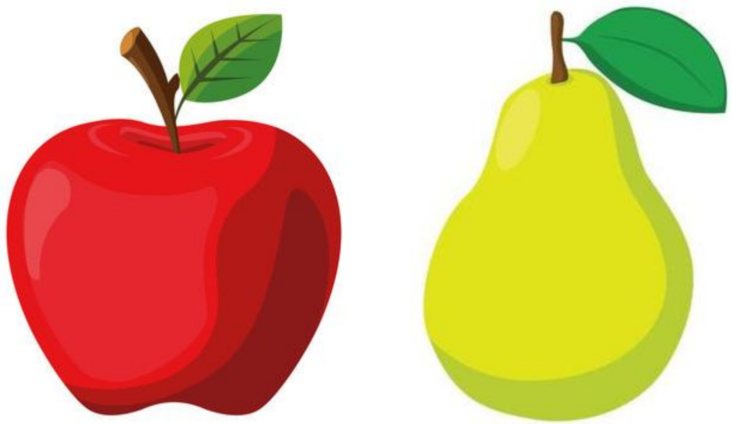
## Week 1 Foundations of Pattern Recognition and Machine Learning

# Foundations of Pattern Recognition

- ✓ Defining Pattern Recognition / Machine Learning
- ✓ Basic Terminologies

# What is Pattern Recognition (PR)?

- Pattern recognition is to **perceive** a pattern, **extract** the relevant info, **understand** the content of the info and **make decision** automatically by machine or computer.
  - In short: automated recognition of patterns and regularities in data.
  - Uses techniques developed in statistics and machine learning.
- Example 1: Is the picture of an apple or pear



- After perceiving the image, you may decide left image is an apple, right image is a pear.
- The decision is made with 100% certainty (probability): how?
- Based on color/shape.

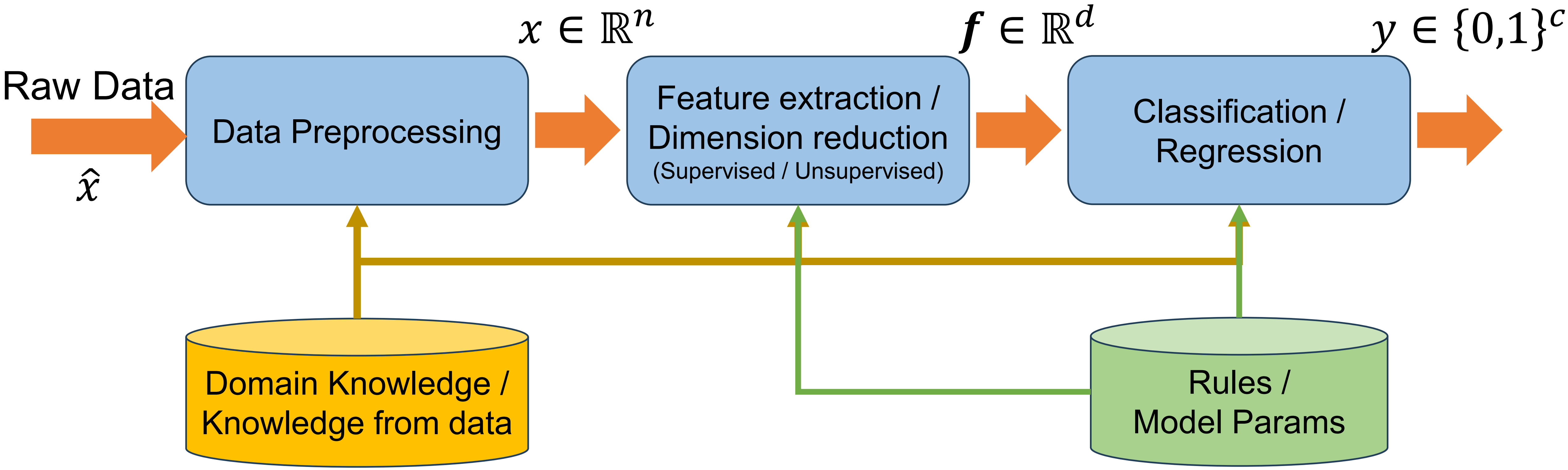
# What is Pattern Recognition?

- Can we always decide with 100% certainty?
  - Example 2: Is the picture of an apple or pear

- Lost of certain information / features (color):
- Can we still make the decision with 100% certainty? Probably not – then how should we determine what the picture is?

# The PR System Diagram

What we usually get here:
01 → apple; 10 → pear;
What do 0 & 1 represent?
Probability!



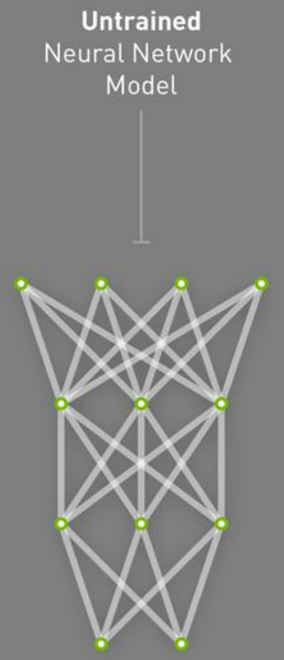
feature

# Pattern Recognition (PR) vs. / and Machine Learning (ML)

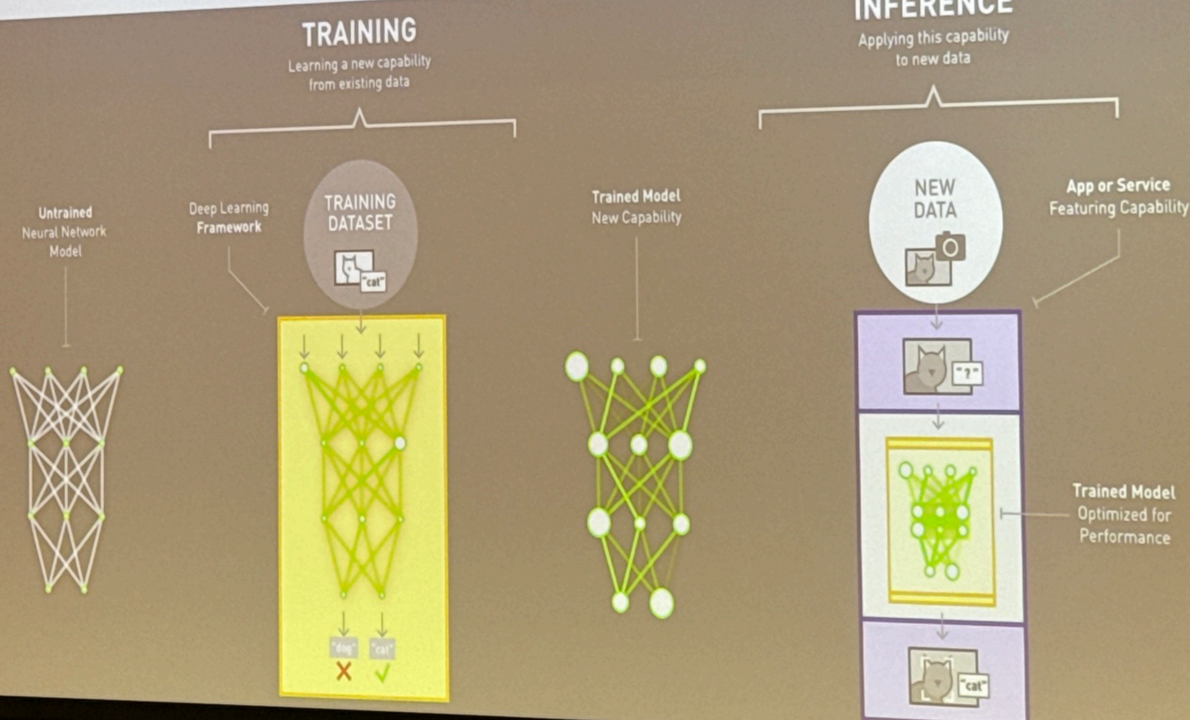
- Connections or differences between PR and ML?
  - Machine learning: learn from data without being explicitly programmed.
  - Almost identical to PR, since learning implies finding patterns.

- What is ML? Compare with non-learning algo (why need ML)?
  - Non-learning algos are based on pre-determined rules, require human expertise. The rules are fixed regardless the input data.
  - For non-learning algos, human expert should know the solution and rules – problems that human cannot solve would NOT be solved.
  - Non-learning algo: human-centric; ML/PR: data-centric.

# The Common ML Framework



# The Common ML Framework

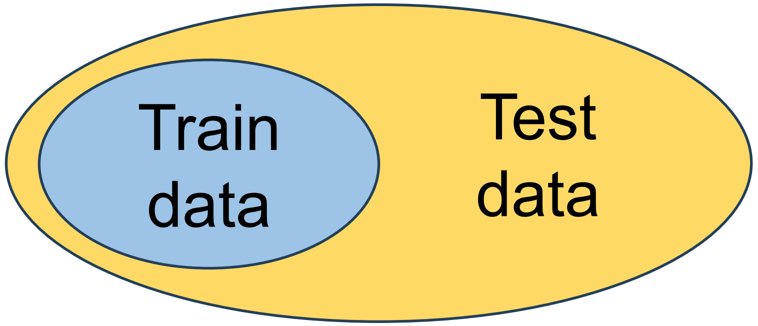


# Inferencing: Deductive / Inductive Reasoning

- Deductive reasoning is a logical approach where you go from **general ideas** to **specific conclusions**.
  - All men are mortal;
  - Socrates is a man;
  - → Socrates is mortal.

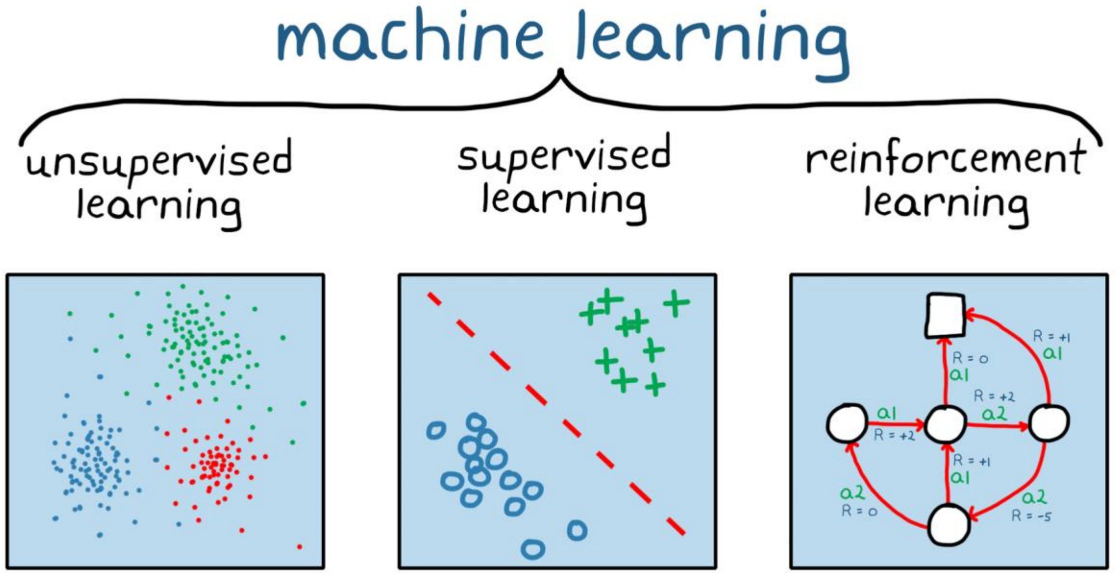
Machine Learning

- Inductive reasoning is a method of drawing conclusion from the **specific** to the **general**.
  - All swans we've seen is white;
  - → All swans are white (?)
- Generalization 泛化



# Different Learning Paradigms

- Supervised learning: Labelled data with guidance;
- Unsupervised learning: No labelled without guidance;
- Reinforcement learning: Interacts with environment, decide action, learns by trial-and-error method.



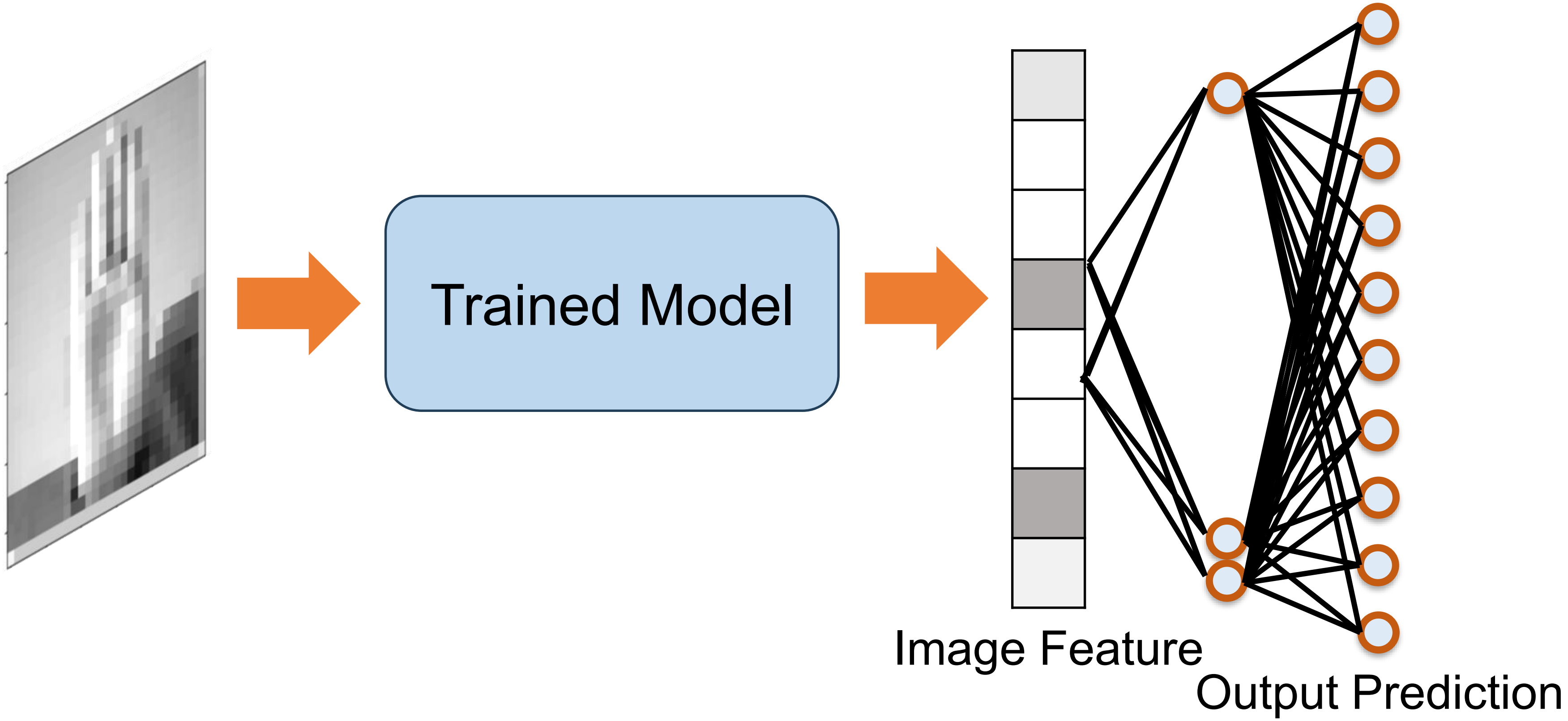
**非监督学习（unsupervised learning，无监督学习）**：训练数据只有输入 $X$，没有人工提供的正确答案 $Y$。

$$\mathcal{D} = \{x_1, x_2, \ldots, x_N\}$$

模型需要自己发现数据内部的结构或规律，例如：

- **聚类（clustering，聚类）**：自动把相似的数据分组。例如根据消费习惯划分用户群体。
- **降维（dimensionality reduction，降维）**：用更少的特征表示数据。例如用 PCA 将高维传感器数据压缩到二维。
- **异常检测（anomaly detection，异常检测）**：找出明显不同于正常数据的样本。例如检测异常电机振动。

# The Common ML Framework: Output




- Note what is the usual output (for classification)?
  - The **probability** of each class!
  - A probabilistic classifier estimates a probability for each class; a decision rule converts these probabilities into a class label.

# Bayesian Reasoning

- ✓ **Introduction to Probability.**
- ✓ **The Bayes' Rule.**
- ✓ **Probabilistic Distributions as Models**

# Probability: Two Interpretations

理解

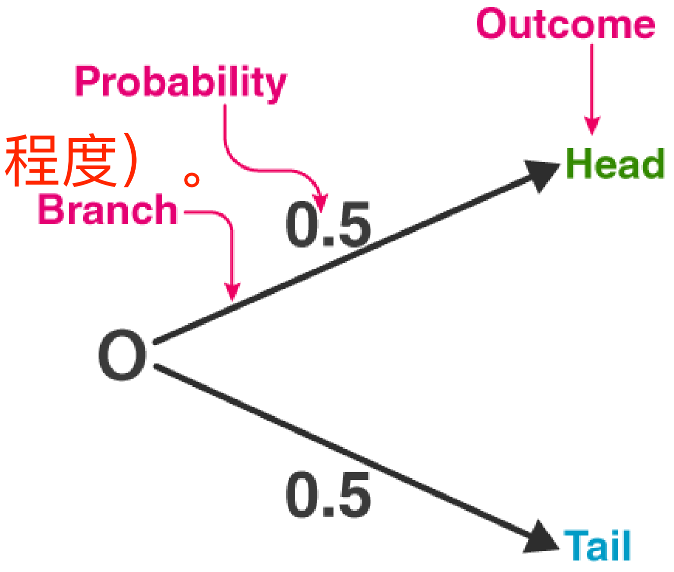
- Probability theory is nothing, but common sense reduced to calculation.
  – Pierre Laplace, 1812

概率表示一个事件经过大量重复实验后出现的长期频率。

- Frequentist interpretation: probabilities represent long run frequencies of events that can happen multiple times.
  - Flip the coin many times, we expect it to land heads about half the time.

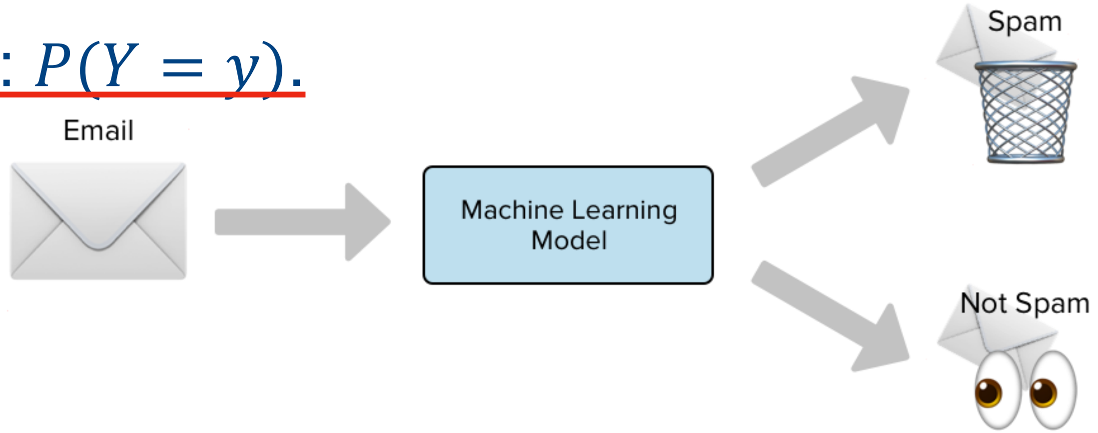
概率表示我们对未知事件的确信程度（degree of belief，信念程度）。

- Bayesian interpretation: quantify uncertainty or ignorance about something
  - The coin is equally likely to land heads/tails on next toss.



# Probability: Events and Random Variables

- Use uppercase letters for random variables (RV): (and in ML)
  - Usually use $X$ as the input / observation / feature vector.
  - Usually use $Y$ as the target label / output variable.
- Use lowercase letters for realized values.
- An event is a statement that a RV takes a particular value: $Y = y$
  - $X = x$: one email's features;
  - $Y = 1$: the email is spam;
  - $Y = 0$: the email is not spam.
- The probability of an event is written as: $P(Y = y)$.



# Properties of Probability of event(s):

- Note: for this course, probability of an event denoted as: $P(A = a)$

- $P(A = a)$ satisfy: $0 \leq P(A = a) \leq 1$.
- Joint probability that events $A$ and $B$ both happening: $P(A = a, B = b)$;
- If $A$ and $B$ are independent events, $P(A = a, B = b) = P(A = a)P(B = b)$.
- Probability of event $A$ or $B$ happening (union of $A = a$ and $B = b$): $P(A = a \vee B = b) = P(A = a) + P(B = b) - P(A = a, B = b)$;
- If $A$ and $B$ are mutually exclusive (cannot happen at the same time), then $P(A = a \vee B = b) = P(A = a) + P(B = b)$;
- Conditional probability of $B = b$ happening given $A = a$ has occurred:

$$P(B = b | A = a) \triangleq \frac{P(A = a, B = b)}{P(A = a)}$$

# Random Variables Map Outcomes to Values

- A mathematical formalization of a quantity or object which depends on random events. 随机变量是对一个“取值由随机事件决定的量”的数学描述。
- A mapping or a function from possible outcomes in a sample space to a measurable space, often the real numbers.
随机变量是一个函数，它把样本空间中的每个可能结果映射到一个可以测量或计算的空间，通常是实数。

| Discrete variable | Continuous variable |
|---|---|
| Countable numbers of value (e.g., number of students) | Infinitely many real values (e.g., time to complete a task) |
| Distribution defined by probability mass function (pmf) | Distribution defined by cumulative distribution function (cdf) and probability density function (pdf) |
| discrete RV for X<br>$p(x) \triangleq P(X = x)$<br>$\sum p(x) = 1$ | $P(x) \triangleq P(X \leq x)$<br>$p(x) \triangleq \frac{d}{dx} P(x)$<br>$P(a < X \leq b) = \int_a^b p(x)dx = P(b) - P(a)$ |

# Discrete and Continuous Random Variables

| Discrete variable | Continuous variable |
|---|---|
| Countable numbers of value (e.g., number of students) | Infinitely many real values (e.g., time to complete a task) |
| Distribution defined by probability mass function (pmf) | Distribution defined by cumulative distribution function (cdf) and probability density function (pdf) |
| $p(x) \triangleq P(X = x)$<br>$\sum p(x) = 1$ | $P(x) \triangleq P(X \leq x)$<br>$p(x) \triangleq \frac{d}{dx} P(x)$<br>$P(a < X \leq b) = \int_a^b p(x)dx = P(b) - P(a)$ |

如何判断一个随机变量是离散的还是连续的。关键不是它表示什么物理量，而是它允许取得哪些值。

- Important: Note whether the variable is discrete or continuous! Could be confusing sometimes. Example:
  - A large quantity of green tea (e.g., 1000kg) is to be sold. $X$ is the weight of green tea to be sold. Suppose any quantity can be sold, is $X$ discrete or continuous?
  - What if the green tea can only be sold in packets of 10kg, is $X$ now discrete or continuous?

# Bayes' Rule: Defining Bayesian

- "Bayesian": refer to inference methods that represent "degrees of certainty" using probability theory, and which leverage Bayes' rule, to update the degree of certainty given data.
  - Update our uncertainty about an unknown hypothesis / class / parameter after observing data.
  - *Layman term: Update our belief based on prior belief and data so that we infer something with a certain degree of uncertainty.*
- Bayes' rule: a formula for computing the probability distribution over possible values of an unknown quantity $Y$ given observed data $X = x$:

$$\boxed{P(Y = y \mid X = x) = \frac{P(X = x \mid Y = y)P(Y = y)}{P(X = x)}}$$

# Bayes' Rule From Conditional Probability

- From conditional probability:

Output Input

$$P(Y = y \mid X = x) = \frac{P(X = x, Y = y)}{P(X = x)}$$

- The joint probability can also be written as:

$$P(X = x, Y = y) = P(X = x \mid Y = y)P(Y = y)$$

- Therefore, we obtain the Bayes' rule:

$$\boxed{P(Y = y \mid X = x) = \frac{P(X = x \mid Y = y)P(Y = y)}{P(X = x)}}$$

normalization
belong to [0,1]

  - $P(Y = y \mid X = x)$: posterior;
  - $P(X = x \mid Y = y)$: likelihood;
  - $P(Y = y)$: prior;
  - $P(X = x)$: evidence.

$$\boxed{\textbf{posterior} = \frac{\textbf{likelihood} \times \textbf{prior}}{\textbf{evidence}}}$$

后验、似然、先验、证据

# Bayes' Rule: in Detail

- $p(Y = y)$: What we know about possible values of $Y$ before we see any data $X = x$
  – The **prior distribution** (prior belief).
- $p(Y = y|X = x)$: After observing data $X = x$, we update this belief to a new belief about the possible values of $Y = y$
  – The **posterior probability** (updated belief).
- $p(X = x|Y = y)$: the distribution over possible outcomes $X = x$ we expect to see if $Y = y$
  – The **likelihood** (the *observed* data).
  – Tell us how compatible the observation $x$ is with a possible label $y$.

- Therefore, Bayesian reasoning can be summarized as:
  - old belief + new evidence → updated belief
- In machine learning, this helps us answer the following question:
  - Given input $x$, how likely is each possible label $y$?

# Bayes' Rule: Example 1 – Weather Forecast

$X = 1$: *Weatherman forecast it rains*
$Y = 1$: *It Actually Rains*

- Marie is getting married tomorrow at an outdoor ceremony in the desert.
- In recent years, it has rained only 5 days each year.

Prior Probability

$$P(Y=1) = \frac{5}{365} = 0.0137$$

- When it actually rains, the weatherman has forecast rain 90% of the time.
- When it doesn't rain, he has forecast rain 10% of the time.

Likelihood

$$P(X=1|Y=1) = 0.9$$
$$P(X=1|Y=0) = 0.1$$

- Unfortunately, the weatherman is forecasting rain for tomorrow.
- What is the probability it will rain on the day of Marie's wedding?

Posterior Probability

$$P(Y=1|X=1) = ??$$

# Bayes' Rule: Example 1 – Weather Forecast

- $P(Y=1)=\frac{5}{365}=0.0137$ and $P(X=1|Y=1)=\ 0.9,\ P(X=1|Y=0)=0.1.$
  What we want: $P(Y=1|X=1).$

$$P(Y=1|X=1)=\frac{P(Y=1)P(X=1|Y=1)}{P(X=1)}=\frac{P(Y=1)P(X=1|Y=1)}{P(X=1|Y=1)P(Y=1)+P(X=1|Y=0)P(Y=0)}$$

$$P(Y=1|X=1)\ =\frac{0.0137(0.9)}{0.9(0.0137)+0.1(1-0.0137)}=0.111$$

- The probability it will rain on the day of Marie's wedding, given the weatherman is forecasting rain for tomorrow is 0.111

# Bayes Rule: Example 2 – Disease Testing

$X = 1$: *Diagnostic test positive*
$Y = 1$: *Jack is infected*

- Suppose there is an outbreak of a viral disease. Jack is taking a diagnostic test to determine if he is infected.
- Suppose the sensitivity (true positive rate) of the diagnostic is 87.5%, and the specificity (true negative rate) of the diagnostic is 97.5%.

Likelihood

$$P(X = 1|Y = 1) = 0.875$$
$$P(X = 0|Y = 0) = 0.975$$

- Also suppose the prevalence of the disease in the area where Jack lives is 10%.

Prior Probability

$$P(Y = 1) = 0.1$$

- Unfortunately, Jack is tested positive. What is the probability of Jack being infected?

Posterior Probability

$$P(Y = 1|X = 1) = ??$$

# Bayes' Rule: Example 2 – Disease Testing

- We know that: $P(Y=1)=0.1$ and $P(X=1|Y=1)=0.875,\ P(X=0|Y=0)=0.975.$
- What we want: $P(Y=1|X=1).$

$$P(Y=1|X=1)=\frac{P(Y=1)P(X=1|Y=1)}{P(X=1)}=\frac{P(Y=1)P(X=1|Y=1)}{P(X=1|Y=1)P(Y=1)+P(X=1|Y=0)P(Y=0)}$$

$$=\frac{P(Y=1)P(X=1|Y=1)}{P(X=1|Y=1)P(Y=1)+\big(1-P(X=0|Y=0)\big)P(Y=0)}$$

$$P(Y=1|X=1)=\frac{P(Y=1)P(X=1|Y=1)}{P(X=1|Y=1)P(X=1)+\big(1-P(X=0|Y=0)\big)\big(1-P(Y=1)\big)}$$

$$=\frac{0.1\times 0.875}{0.875\times 0.1+(1-0.1)(1-0.975)}=0.795$$

- The chance of Jack being infected is 79.5%.

# Bayes Rule: Example 2 – Disease Testing

- Suppose there is an outbreak of a viral disease. Jack is taking a diagnostic test to determine if he is infected.
- Suppose the sensitivity (true positive rate) of the diagnostic is 87.5%, and the specificity (true negative rate) of the diagnostic is 97.5%.
- Now suppose the prevalence of the disease in the area where Jack lives is changed to 1%.
- Jack is tested positive. What is the probability of Jack being infected now?

$X = 1$: *Diagnostic test positive*
$Y = 1$: *Jack is infected*

Likelihood

$$P(X = 1|Y = 1) = 0.875$$
$$P(X = 0|Y = 0) = 0.975$$

Prior Probability

$$P(Y = 1) = 0.01$$

Posterior Probability

$$P(Y = 1|X = 1) = ??$$

# Bayes' Rule: Example 2 – Disease Testing

- We know that: $P(Y=1)=0.01$ and $P(X=1|Y=1)=0.875,\ P(X=0|Y=0)=0.975.$
- What we want: $P(Y=1|X=1).$

$$P(Y=1|X=1)=\frac{P(Y=1)P(X=1|Y=1)}{P(X=1)}=\frac{P(Y=1)P(X=1|Y=1)}{P(X=1|Y=1)P(Y=1)+P(X=1|Y=0)P(Y=0)}$$

$$=\frac{P(Y=1)P(X=1|Y=1)}{P(X=1|Y=1)P(Y=1)+\big(1-P(X=0|Y=0)\big)P(Y=0)}$$

$$P(Y=1|X=1)=\frac{P(Y=1)P(X=1|Y=1)}{P(X=1|Y=1)P(X=1)+\big(1-P(X=0|Y=0)\big)\big(1-P(Y=1)\big)}$$

$$=\frac{0.01\times 0.875}{0.875\times 0.01+(1-0.01)(1-0.975)}=\cancel{0.795}\quad 0.261$$

- The chance of Jack being infected is 26.1%, even if tested positive.

# Lessons from Bayes' Rule

- Probability is not only for counting outcomes.
- It is also a way to update uncertainty and beliefs.
- In machine learning, we use this idea to reason about:
  - Labels;
  - Model parameters;
  - Predictions;
  - Uncertainty.

- With this understanding, we move from probability as pure "calculation" to understanding probability as "models for understanding data".

# Summary: Bayes' Rule Updates Beliefs with Evidence

- Let $Y$ be an unknown hypothesis, class or state, and $X$ the observed evidence.
  - **Prior** $p(Y)$**:** our belief about $Y$ before observing $X$.
  - **Likelihood** $p(X \mid Y)$**:** how compatible the observed $X$ is with each possible hypothesis $Y$.
  - **Prior** $\times$ **likelihood:** combines what we believed with what the new evidence tells us.
  - **Evidence** $p(X)$**:** normalizes the result so that the posterior probabilities sum to one.
  - **Posterior** $p(Y \mid X)$**:** our updated belief about $Y$ after observing $X$.
    - **Posterior belief ∝ Likelihood × Prior belief**
  - In sequential learning, the posterior obtained from the current data can become the prior when new data arrive.
- Bayes' rule updates probabilities; choosing an action or class label additionally requires a **decision rule and a loss function**. (Next week)

# Probabilistic Models and Distribution as Models

- In Bayesian/probabilistic learning:
  - **"A model is a probability distribution"**
  - The distribution describes how likely different outcomes are.

- Different prediction targets require different distributions!

| ML task | Target $(y)$ | Probability model |
|---|---|---|
| Binary classification | $(0/1)$ | **Bernoulli**, Binomial |
| Multi-class classification | $(1, \ldots, C)$ | **Categorical**, Multinomial |
| Regression | Real value | **Gaussian** |

# Bernoulli Distribution

伯努利

- One of the simplest probability distribution – can be used to model binary events. (E.g., coin flipping)
  - Denote $y = 1$ as coin lands on heads, and $y = 0$ as coin lands on tails. Suppose the probability of the coin lands on heads is $\theta \in [0,1]$, $P(y = 1) = \theta$ and $P(y = 0) = 1 - \theta$. Then random variable $Y$ follow the Bernoulli distribution: $y \sim \mathrm{Ber}(\theta)$.

$$\boxed{\mathrm{Ber}(y|\theta) = \begin{cases} 1 - \theta & if\ y = 0 \\ \theta & if\ y = 1 \end{cases} \triangleq \theta^y (1-\theta)^{1-y}, \text{or } P(y|\theta) = \theta^y (1-\theta)^{1-y}}$$

  - For classification, probability depends on the input formulated as: $P(y|x,\theta)$
  - Example: $P(y = 1|x,\theta) = 0.8$ means that the model predicts 80% probability of the coin landing on heads.
  - Final class prediction can be made by thresholding: (not just for binary, actually)

$$\hat{y} = 1 \quad \text{if } P(y = 1|x,\theta) > \mathrm{Thres}$$

# Binomial Distribution

二项

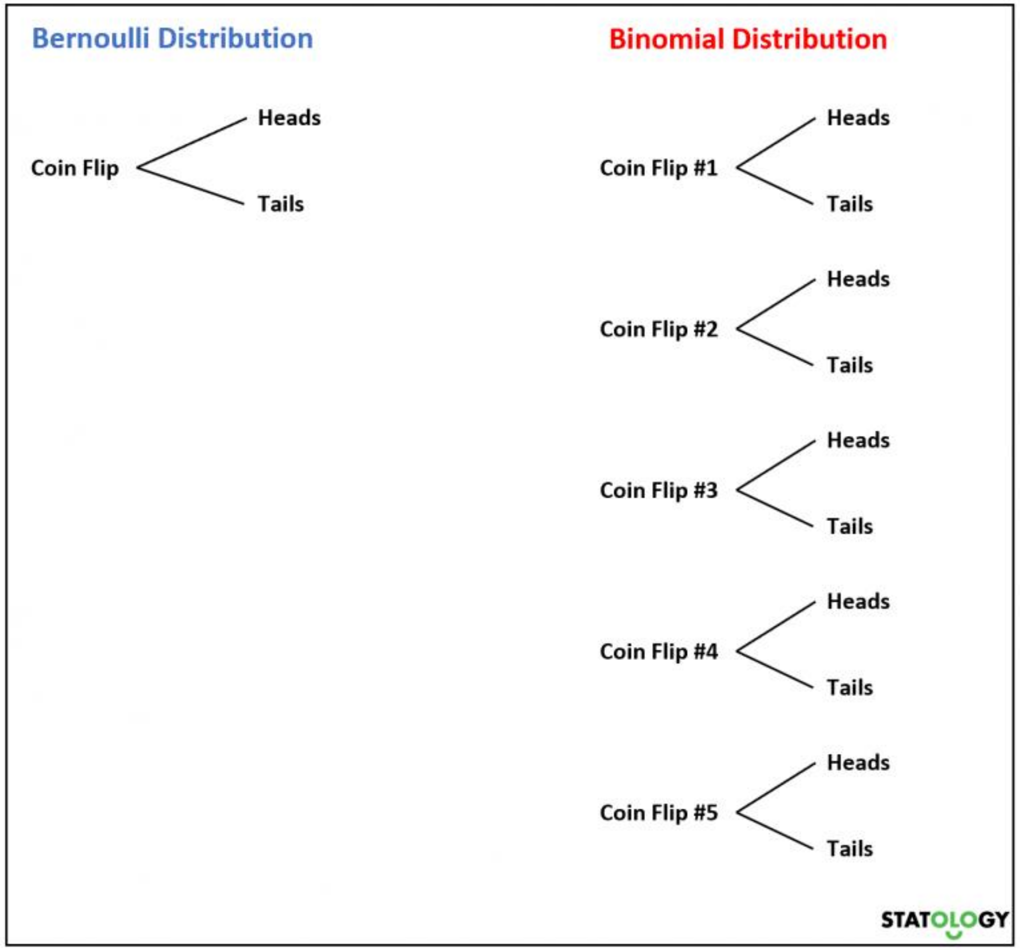
- Now, if we flip a coin multiple times. Then the sum of the Bernoulli random variables will follow a **Binomial** distribution.
  - The Bernoulli distribution is a **special case** of the binomial distribution.

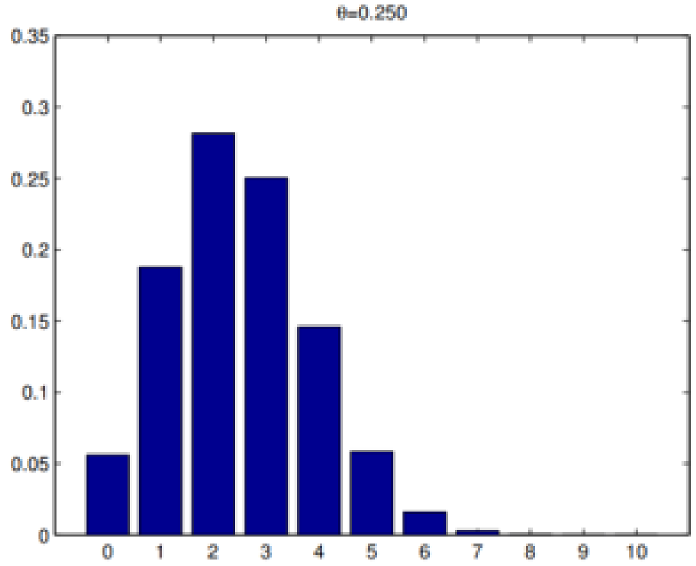
  - Suppose we observe a set of $N$ Bernoulli trials (e.g., toss a coin $N$ times), $y_n \sim \mathrm{Ber}(\cdot|\theta)$ for $n = 1:N$. Define $s$ to be the total number of heads $s \triangleq \sum_{n=1}^{N} \mathbb{I}(y_n = 1)$.
  - The distribution of $s$ follows the binomial distribution:

$$\mathrm{Bin}(s|N,\theta) \triangleq \binom{N}{s} \theta^s (1-\theta)^{N-s} = \boldsymbol{C}_s^N \theta^s (1-\theta)^{N-s}$$

组合数 binomial coefficient: $\binom{N}{s} \triangleq \frac{N!}{(N-s)!\,s!}$

# Binomial Distribution

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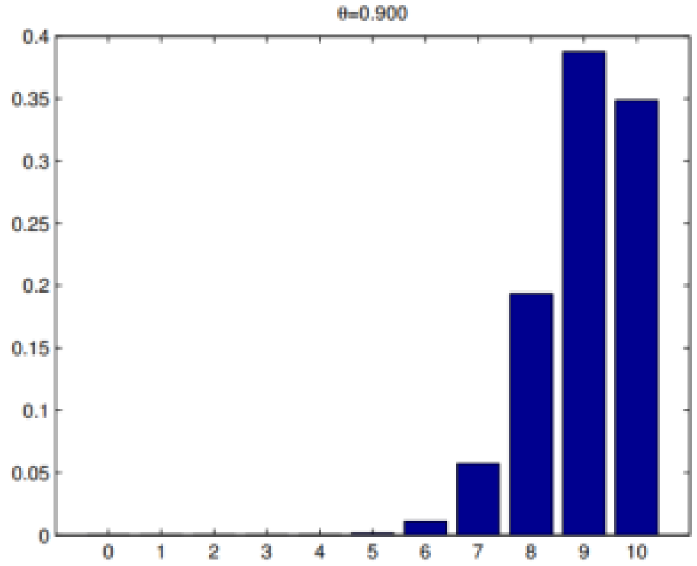


(b)

# End of Week 1
# Thank you!